Detection and Contouring of Cancerous Tissue in Lung Cancer Patients Using Artificial Intelligence Methods

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Abstract— **This study investigates the use of deep learning models for processing and analyzing PET/CT (Positron Emission Tomography/Computed Tomography) images in the diagnosis of lung cancer. Lung cancer is one of the most common and fatal types of cancer worldwide, and early diagnosis and accurate treatment methods can significantly improve the prognosis of the disease. PET/CT imaging is widely used for the detection and evaluation of cancerous tissues. However, manual analysis of these images is time-consuming and prone to errors. In this study, deep learning models capable of high-accuracy and high-sensitivity classification and segmentation in lung cancer diagnosis were developed using PET/CT images obtained from the Department of Nuclear Medicine at Akdeniz University Hospital. Advanced deep learning models were employed to automatically differentiate between cancerous and healthy tissues and precisely contour cancerous regions. The results demonstrate that deep learning models can be used with high accuracy in lung cancer diagnosis. This study emphasizes the effectiveness of deep learning techniques in processing and analyzing PET/CT images and opens new horizons in medical image analysis.**

Keywords: **Lung cancer, PET/CT imaging, Deep learning, Classification, Segmentation**

INTRODUCTION

Cancer emerges as a serious health issue globally (1). According to the latest global cancer statistics, 14.1 million new cancer cases have developed, and 8.2 million deaths have resulted from cancer. If the current trend continues, 22 million new cases are expected by 2030 (2). As shown in Figure 1, lung cancer ranks first among cancer cases worldwide in terms of both incidence and mortality rates. Early diagnosis and accurate treatment planning can significantly increase patients' lifespan and quality of life. In this study, we investigated the use of artificial intelligence (AI)-assisted image processing techniques in the diagnosis and treatment planning of lung cancer. Our goal is to contribute to the clinical diagnosis process by automatically detecting and contouring cancerous tissues from PET/CT images.

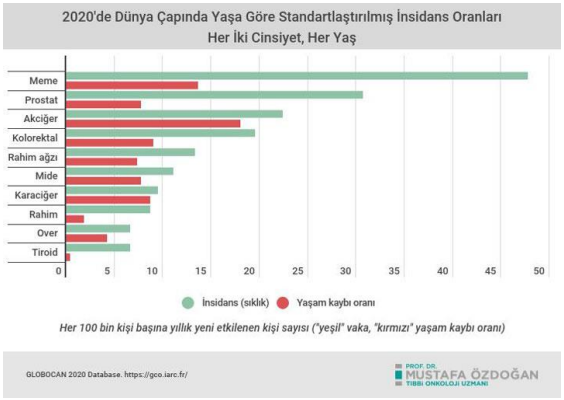


Figure 1. Global Cancer Data

In this study, a two-stage approach was developed using PET/CT images of patients diagnosed with lung cancer. In the first stage, a classification model based on InceptionV3 was trained to automatically distinguish between cancerous and healthy images. Sections identified as cancerous by the classification model were sent to a YOLOv8-based segmentation model for precise contouring. During this process, the segmentation model automatically contoured the cancerous regions.

The main steps of the project are as follows:

1. **Data Preparation:** DICOM files obtained from PET/CT images were converted to PNG format after certain preprocessing steps. During this stage, DICOM files were cleaned and processed in a specific sequence.
2. **Model Training:** The classification model was trained with training data. The model successfully distinguishes between cancerous and healthy images and directs the detected cancerous sections to the segmentation model.
3. **Segmentation:** The segmentation model precisely contours the cancerous sections identified by the classification model. During this stage, segmentation results were visualized, and the accuracy of the model was evaluated by comparing them with real masks.
4. **Imaging and Evaluation:** The developed system presents the results to the user through a GUI (Graphical User Interface) and visualizes DICOM images. Additionally, the performance

of the system was evaluated using various metrics.

LITERATURE REVIEW

Machine learning and deep learning techniques have made significant advancements in medical imaging and cancer diagnosis in recent years. Models developed for the diagnosis and segmentation of lung cancer have the potential to improve patient management by providing accurate and rapid diagnosis. In this section, studies on lung cancer classification and segmentation in the literature are reviewed.

Lung Cancer Classification

Lung cancer classification studies generally rely on computer vision and deep learning techniques. Deep learning models, especially convolutional neural networks (CNNs) (Figure 2), are widely used in processing medical images.

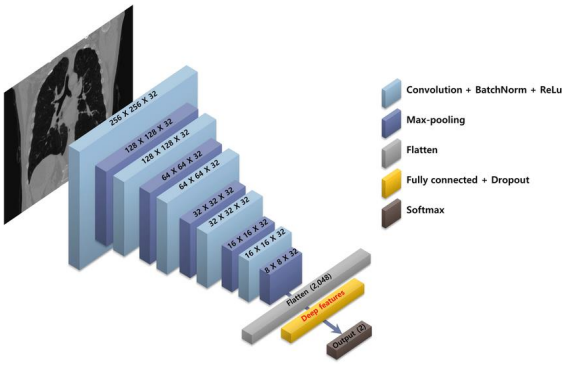


Figure 2. An example of CNN architecture

Ardila et al. (2019) performed lung cancer screening using a three-dimensional deep learning model on low-dose chest tomography (LDCT) images. The model showed high performance with a 94% accuracy rate (3). Similarly, Nam et al. (2019) developed a deep learning-based automatic detection algorithm and used it for detecting malignant pulmonary nodules in chest radiographs, achieving a high success rate with a 92% accuracy (4). This study demonstrates that deep learning algorithms can be a reliable method for cancer screening.

Sevli and Pazar (2020) worked on the classification of lung cancer data using machine learning methods. In the study, various machine learning algorithms such as support vector machines (SVMs) (Figure 3), random forest, and logistic regression were used, with logistic regression showing the highest performance with 97% accuracy (5). Additionally, Zhang et al. (2020) achieved 93.1% accuracy using a ResNet50-based model (6).

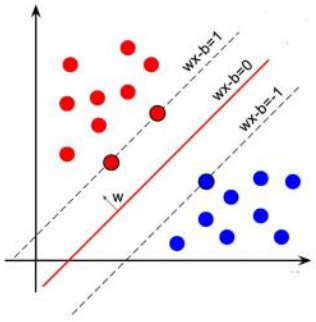


Figure 3. Support Vector Machine

Lung Cancer Segmentation

Lung cancer segmentation studies aim to accurately identify and contour cancerous regions. Segmentation techniques are used to determine the volume and spread of cancerous tissues accurately.

Çakar et al. (2017) applied image processing techniques to detect candidate nodules in lung tomographies using data from the National Cancer Institute (NCI). In this study, candidate nodules were detected using Otsu thresholding (Figure 4) and laplacian filtering methods (7). Simões et al. (2019) performed geometric and dosimetric evaluations of automatic contouring methods for breast cancer target volume and showed that automatic contouring provides results comparable to manual contouring (8).

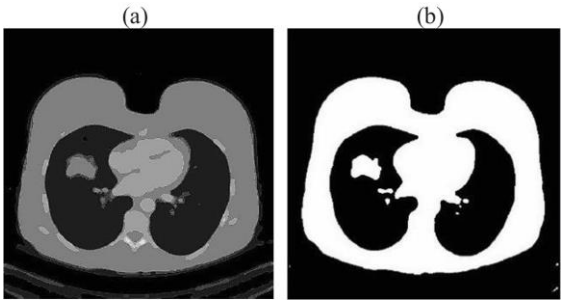


Figure 4. (a) Filtered Image (b) Image Processed with Otsu Thresholding Method

MATERIAL AND METHODS

Data Set

In this study, PET/CT images of patients diagnosed with lung cancer from the Department of Nuclear Medicine at Akdeniz University Hospital were used. The dataset consists of DICOM format images obtained from various patients, containing 3000 cancerous and 3000 healthy sections.

The data preparation process consists of several steps to make the data usable for classification and segmentation models:

- First, DICOM files were converted to PNG format and filtered in a specific range. Converted PNG files were saved in a specific directory for model training and evaluation.
- The data were prepared in a suitable format for training classification and segmentation models; images separated into training, validation, and test sets were normalized and scaled to a specific size.
- During the data preparation stage of the developed system, cancerous regions were marked using QuPath software, and annotations of these regions were extracted. Annotations were saved in JSON format, various preprocessing steps were performed on these data, and they were used in training the segmentation model.
- Masks were created from the obtained annotations and saved in a specific directory, and these masks were used for training the segmentation model.

Imaging and Evaluation

Visualizing and evaluating the data is crucial for determining the model's performance. In this study, the created masks and real masks were visualized and compared. Matplotlib and OpenCV libraries were used for visualization. The overlap ratio between the predicted masks and real masks was calculated using the IoU metric. Additionally, the converted PNG files were reverted to DICOM format without any data loss after the model training and contouring processes. The developed GUI (Figure 5) enabled users to view DICOM images and examine the processed results. The DICOM viewer allowed detailed examination of the images. These methods provided a detailed evaluation of the model performance.

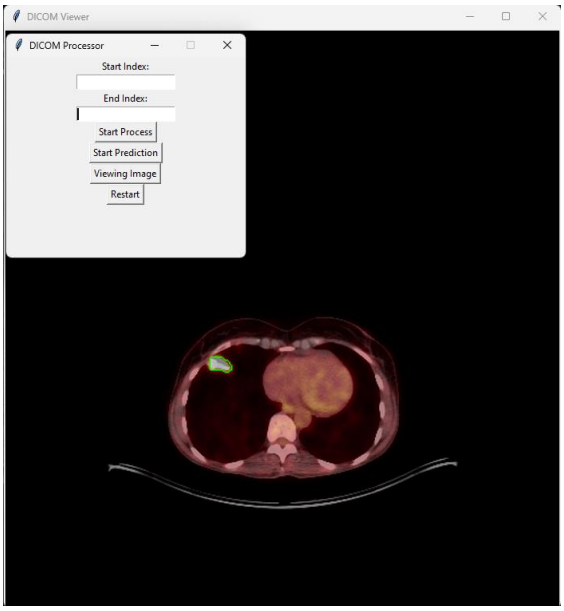


Figure 5. GUI

Machine Learning Techniques Used

In this study, various machine learning techniques were used for the classification and segmentation of lung cancer images. An InceptionV3-based classification model and a YOLOv8-based segmentation model were preferred as the main components.

The InceptionV3 model used as the classification model is a deep learning model developed by Google and is commonly used for image classification tasks. In this study, the InceptionV3 model was used to classify lung cancer images as cancerous and healthy. The model performs multiple convolution operations in the same layer using filters of different sizes to learn features of different sizes simultaneously. The model, initialized with pre-trained weights on the ImageNet dataset, was re-trained with lung cancer images. The model was compiled using the Adam optimization algorithm, and categorical cross-entropy loss function and accuracy metric were used throughout the training process.

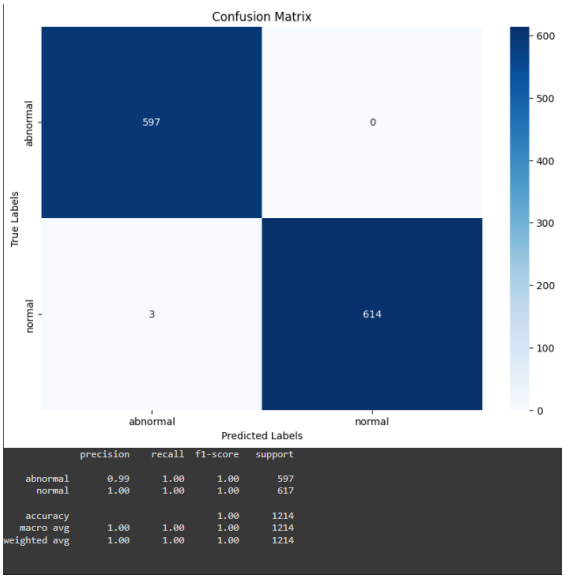
YOLOv8 was used as the segmentation model. The YOLO (You Only Look Once) algorithm is a deep learning model used for real-time object detection and segmentation, and YOLOv8 is the latest and most advanced version of this algorithm. YOLOv8 has an architecture that can perform faster and more accurate object detection and segmentation. The model was trained to precisely contour the cancerous sections identified by the classification model. The SGD optimization algorithm was used during the training process.

EXPERIMENTAL RESULTS

In this section, the experimental results evaluating the performance of the classification and segmentation models will be presented and analyzed. Performance metrics, visualization results, and comparative analyses of the models on training and validation sets will be detailed.

Classification Model Results

Performance metrics such as accuracy, precision, specificity, and F1 score of the model on the training and validation sets are presented in Figure 6.



	precision	recall	f1-score	support
abnormal	0.99	1.00	1.00	597
normal	1.00	1.00	1.00	617
accuracy			1.00	1214
macro avg	1.00	1.00	1.00	1214
weighted avg	1.00	1.00	1.00	1214

Figure 6. Performance Metrics of the Classification Model

The confusion matrix and performance metrics table evaluate the model's performance in diagnosing lung cancer. The confusion matrix shows the relationship between the labels predicted by the model and the actual labels. The four main regions in the matrix are as follows:

- True Positive (TP): Samples correctly predicted as "abnormal" by the model (597)
- True Negative (TN): Samples correctly predicted as "normal" by the model (614)
- False Positive (FP): Samples predicted as "abnormal" by the model but actually "normal" (0)
- False Negative (FN): Samples predicted as "normal" by the model but actually "abnormal" (3)

Performance metrics are used to evaluate the overall success of the model. Below are the definitions of the metrics and their interpretations for this model:

- **Precision:** Shows how many of the samples predicted as "abnormal" by the model are actually "abnormal." The precision for the abnormal class is 0.99, and for the normal class, it is 1.00. These high precision values indicate that the model makes accurate predictions and the false positive rate is very low.
- **Recall:** Shows how many of the samples that are actually "abnormal" are correctly predicted as "abnormal." The recall for both the abnormal and normal classes is 1.00. These values indicate that the model has almost no false negatives and shows high sensitivity.
- **F1-Score:** The harmonic mean of precision and recall. It is useful when both precision and recall are balanced. The F1-score value for both the abnormal and normal classes is 1.00. This indicates that the model shows perfect performance for both classes.
- **Accuracy:** The ratio of correctly predicted samples to the total number of samples and was obtained as 100% in this study. This result indicates that the model classifies all samples correctly and its overall performance is excellent.
- **Support:** The support metric, showing the actual number of samples for each class, is recorded as 597 for the abnormal class and 617 for the normal class. These data indicate that the model was trained and tested with a sufficient amount of data.

The confusion matrix and performance metrics of the InceptionV3 model reveal that the model shows very high performance in diagnosing lung cancer. The model achieved an accuracy rate of 100%, correctly classifying almost all "abnormal" and "normal" samples (Figure 7). Especially, the precision and recall values are both 1.00 or close to it, indicating that the model maintains high true positive and true negative rates. Only three "normal" samples were incorrectly classified as "abnormal," which does not significantly affect the model's performance.

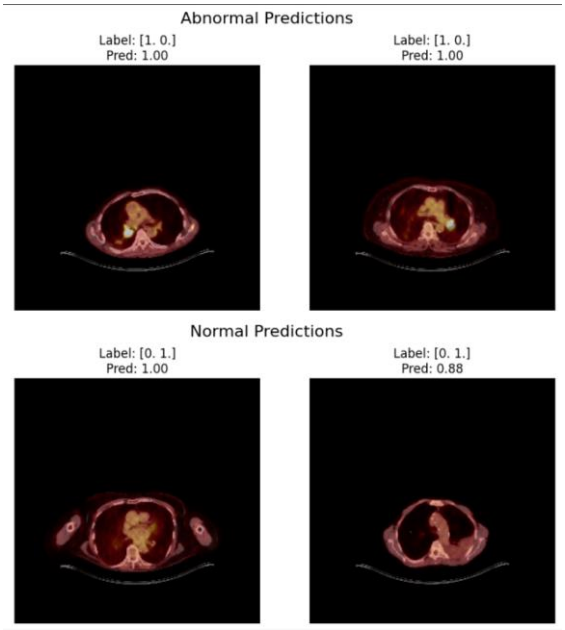


Figure 7. Predictions of the Classification Model

These results are competitive and show superior performance compared to other studies on lung cancer image classification in the literature. For example, Zhang et al. (2020) achieved 93.1% accuracy using a ResNet50-based model (6). The InceptionV3 model used in this study outperformed this study with an accuracy value of 100%. Additionally, Hassani et al. (2019) achieved 91.5% accuracy with a VGG16-based model (9). The developed model shows higher performance than this study as well, providing a more reliable classification in lung cancer diagnosis.

Training and Validation Loss and Accuracy Values

The training and validation loss and accuracy values of the model during the training process are shown in Figures 8 and 9. The decrease in the loss function and the increase in accuracy values during the training process indicate that the model performed well during the learning process.

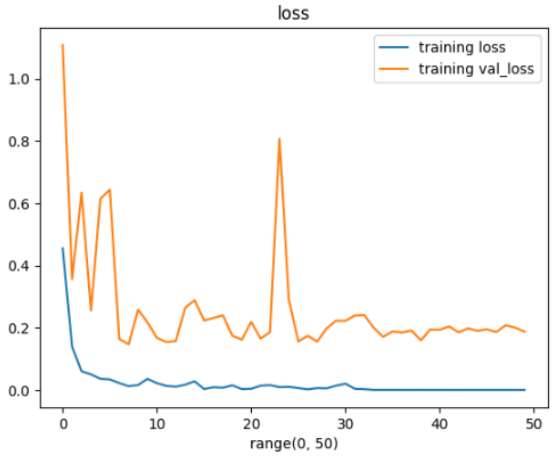


Figure 8. Training and Validation Loss Graph

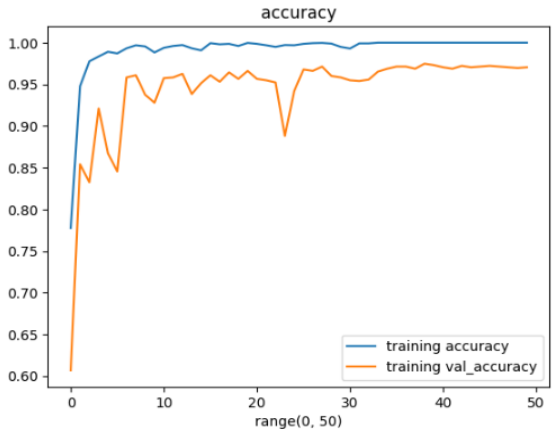


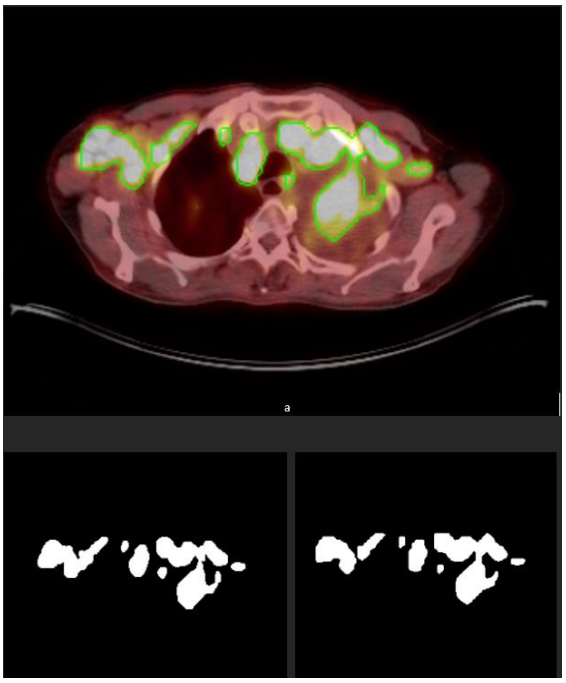
Figure 9. Training and Validation Accuracy Graph

Segmentation Model Results

In this section, the performance metrics of the segmentation model will be evaluated and discussed.

Visualization of Predicted and Real Masks

For randomly selected images, the predicted masks and real masks were visualized and compared. In the figure, (a) shows the predicted contours, (b) shows the real masks, and (c) shows the predicted masks. It is observed how well the predicted masks overlap with the real masks.



(b) *(c)*

Additionally, the IoU value for the predicted masks was calculated, and the segmentation performance of the model was evaluated.

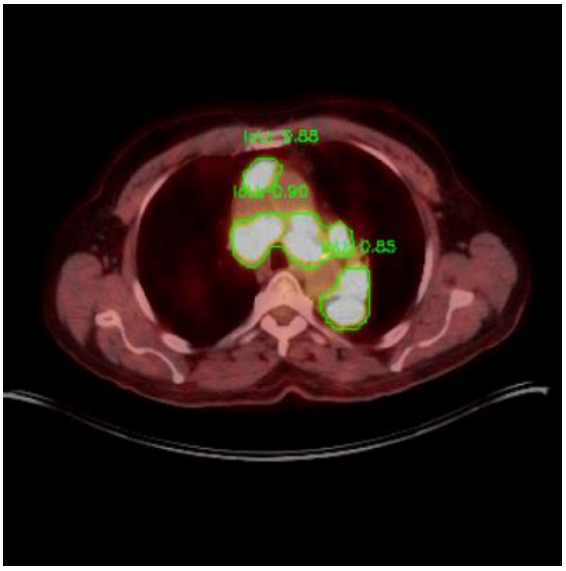


Figure 11. IoU Values of Contours

Performance Metrics:

- **Precision:** Precision shows how many of the segmentations identified by the model are correct. The precision for benign (non-malignant) samples is 0.888, and for malignant samples, it is 0.824. These values indicate that the model works with high accuracy, especially for benign samples. Although the precision for malignant samples is slightly lower, it is still acceptable.
- **Recall:** Recall shows how well the model detects the actual correct segmentations. The recall for benign samples is 0.817, and for malignant samples, it is 0.767. This indicates that the model has a high detection rate, especially for benign samples, but the detection rate for malignant samples is slightly lower.
- **mAP50:** The mean average precision for segmentations with an IoU threshold of 0.50. The mAP50 value for benign samples is 0.897, and for malignant samples, it is 0.828. This indicates that the segmentation accuracy of the model is high.

DISCUSSION AND CONCLUSION

In this section, the findings of the study and the obtained results will be discussed, the strengths and weaknesses of the study will be addressed, and suggestions for future studies will be made.

Discussion of Findings

In this study, the performance of the InceptionV3-based classification model and the YOLOv8-based segmentation model developed for use in the diagnosis and treatment planning of lung cancer was evaluated. The obtained results demonstrate that the developed models can successfully classify lung cancer images and precisely contour cancerous regions.

- **Classification Performance:** The InceptionV3-based classification model achieved high accuracy, precision, and specificity values. This indicates that the model can effectively distinguish between cancerous and healthy images. However, the false positive and false negative rates should also be considered, and to reduce these rates, the model can be trained with more data or more complex models can be used.
- **Segmentation Performance:** The YOLOv8-based segmentation model could accurately contour cancerous regions. This increases the usability of the model in clinical applications. However, some images obtained low IoU values, indicating that the model is less successful in certain situations. To resolve this issue, the model should be trained with more data, and segmentation algorithms should be improved.

Strengths and Weaknesses

Strengths:

- **High Accuracy and Precision:** The developed models can be used in lung cancer diagnosis and treatment with high accuracy and precision values.
- **Comprehensive Data Preparation:** Detailed data preparation steps such as cleaning DICOM files, converting them to PNG format, and extracting annotations positively impacted model performance.
- **Patient Data:** During the conversion of DICOM files to PNG format, it was ensured that no patient data in the DICOM files were damaged.

Weaknesses:

- **Limited Data Set:** The used data set consists of a limited number of images obtained from a specific hospital. Model performance can be increased with larger and more diverse data sets.
- **False Positive and Negative Rates:** There are false positive and negative rates in the classification model, and these rates need to be reduced.
- **Segmentation Errors:** Some images obtained segmentation errors and low IoU values, indicating that the model is less successful in certain situations.

Suggestions for Future Studies

- **Larger and More Diverse Data Sets:** To increase model performance, larger and more diverse data sets should be used. Data obtained from different hospitals and different patient groups can increase the generalization ability of the model.
- **Model Improvements:** More complex models and advanced data augmentation techniques can be used to reduce false positive and negative rates. Additionally, new algorithms and optimization techniques can be researched to improve the performance of the segmentation model.
- **Clinical Applications:** Comprehensive clinical evaluations should be conducted for the use of developed models in clinical applications. The model should be tested with real-world data and improved with clinical feedback.
- **Integrated Systems:** By developing an integrated system combining classification and segmentation models, clinical diagnosis processes can be made more efficient and effective. This system can help doctors make quick and accurate decisions.

Conclusion

This study demonstrates that AI-assisted classification and segmentation techniques can be effectively used in the diagnosis and treatment planning of lung cancer. These findings highlight the potential of AI technologies in the healthcare field and provide a significant foundation for future studies. In the future, the performance of these systems can be further enhanced with larger data sets and advanced techniques, and their widespread use in clinical applications is targeted.

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